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EXPERIMENT 15

Transformation using Direct Linear Transformation (DLT)

AIM

To estimate a homography from point correspondences using the Direct Linear Transformation algorithm, and to verify it against cv2.findHomography().

PROGRAM

```
import cv2
import numpy as np

def dlt_homography(src, dst):
    """Estimate a 3x3 homography from >= 4 point correspondences using DLT.

    Each correspondence (x, y) -> (u, v) contributes two rows to A.
    The homography is the null-space vector of A, i.e. the right
    singular vector of A with the smallest singular value.
    """
    A = []
    for (x, y), (u, v) in zip(src, dst):
        A.append([-x, -y, -1, 0, 0, 0, u * x, u * y, u])
        A.append([0, 0, 0, -x, -y, -1, v * x, v * y, v])
    A = np.array(A, dtype=np.float64)

    _, _, Vt = np.linalg.svd(A)
    H = Vt[-1].reshape(3, 3)
    return H / H[2, 2] # normalise so that H[2][2] = 1

img = cv2.imread("image1.jpg")
rows, cols = img.shape[:2]

src_points = np.float32([[0, 0], [cols - 1, 0],
                        [0, rows - 1], [cols - 1, rows - 1]])
dst_points = np.float32([[60, 40], [cols - 30, 15],
                        [10, rows - 20], [cols - 70, rows - 60]])

H = dlt_homography(src_points, dst_points)
dlt_img = cv2.warpPerspective(img, H, (cols, rows))

print("Homography by DLT:\n", H)
H_cv, _ = cv2.findHomography(src_points, dst_points)
print("Homography by cv2.findHomography():\n", H_cv)
print("Max difference:", np.abs(H - H_cv).max())

cv2.imwrite("dlt_transformed.jpg", dlt_img)
cv2.imshow("Original", img)
cv2.imshow("DLT Transformed", dlt_img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT



image1.jpg (input image)

OUTPUT



dlt_transformed.jpg (warped using the homography computed by the DLT algorithm)

Console output:

```
Homography by DLT:
[[ 8.56469744e-01 -9.82414992e-02  6.00000000e+01]
 [-4.79702131e-02  8.65148375e-01  4.00000000e+01]
 [ 6.35644005e-05 -3.94141102e-05  1.00000000e+00]]
Homography by cv2.findHomography():
[[ 8.56469744e-01 -9.82414992e-02  6.00000000e+01]
 [-4.79702131e-02  8.65148375e-01  4.00000000e+01]
 [ 6.35644005e-05 -3.94141102e-05  1.00000000e+00]]
Max difference: 1.0914718018284475e-09
```

RESULT

Thus a homography was successfully estimated using the Direct Linear Transformation algorithm, and the matrix obtained matches the one returned by `cv2.findHomography()`.